Date: 20-1-2025 Lec # 04 Day: M T W T F S

Limit & Continuity

i) limits of two variable.

Exp :- evalute $\lim_{(x,y)\to(1,2)} (x^3 + 3xy - 2y^3)$

↓ Polynomial func.

↓

In P.F direct limit put.

Solution:-

Direct Substitution

$$\lim_{(x,y)\to(1,2)} x^3 + 3xy - 2y^3$$

$$= (1)^3 + 3(1)(2) - 2(2)^3$$

$$= 1 + 3(2) - 2(8)$$

$$= 1 + 6 - 16$$

$$= \boxed{-9} \text{ Ans.}$$

Exp : 2 $\lim_{(x,y)\to(5,5)} \dfrac{x^2 - y^2}{x - y}$.

Solution:-

by putting direct limit, answer will be undefined. $\boxed{\dfrac{0}{0}}$

Factorization method

$$a^2 - b^2 = (a-b)(a+b)$$

$$\lim_{(x,y)\to(5,5)} \frac{x^2 - y^2}{x - y}$$

$$\frac{\cancel{(x-y)}(x+y)}{\cancel{(x-y)}}$$

$$= (x+y)$$

$$= 5 + 5$$

$$= \boxed{10} \text{ Ans.}$$

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(B) $\lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+y^2}$

limit exist ?

Solution:-

$$\lim_{(x,y)\to(0,0)} \boxed{\frac{0}{0}}$$

along x-axis, $y = 0$

$$\lim_{(x,0)} \frac{x^2}{x^2+y^2}$$

$$= \frac{x^2}{x^2+0^2}$$

$$\lim_{(x,0)} = \frac{x^2}{x^2} = 1$$

$$\boxed{\lim_{(x,0)} \frac{x^2}{x^2+y^2} = 1}$$

along y-axis, $x = 0$

$$\lim_{(0,y)} \frac{x^2}{x^2+y^2}$$

$$= \frac{(0)^2}{(0)^2+y^2} = 0$$

$$\boxed{\lim_{(0,y)} \frac{x^2}{x^2+y^2} = 0}$$

limit doesn't exists because $f(x,y)$ has two different limits along two different lines

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④ $$\lim_{(x,y)\to(0,0)} \frac{x^2y^2}{x^4+y^4}$$

limit exist ?

Solution:-

$$\lim_{(x,y)\to(0,0)} \left(\frac{0}{0}\right).$$

along x-axis, $y=0$

$$\lim_{(x,0)} \frac{x^2y^2}{x^4+y^4}$$

$$= \frac{x^2(0)^2}{x^4+(0)^4} = \frac{0}{x^4} = 0.$$

$$\boxed{\lim_{(x,0)} \frac{x^2y^2}{x^4+y^4} = 0}$$

along y-axis, $x=0$

$$\lim_{(0,y)} \frac{x^2y^2}{x^4+y^4}$$

$$= \frac{(0)^2\,y^2}{(0)^4+y^4} = \frac{0}{0} = 0$$

$$\boxed{\lim_{(0,y)} = \frac{x^2y^2}{x^4+y^4} = 0}$$

Now we take another line, we say $y=x$

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$$\lim_{(x,y)\to(0,0)} \frac{x^2 y^2}{x^4 + y^4}$$

$$\lim_{(x,x)\to(0,0)} \frac{x^2 (x)^2}{x^4 + (x)^4} = \frac{x^4}{2x^4} = \boxed{\frac{1}{2}}$$

limit doesn't exists, b/c limit of $f(x,y)$ is different at different lines.

(5) $$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2 + y^2}$$

Solution:-

along x-axis, $y = 0$

$$\lim_{(x,0)} \frac{xy^2}{x^2 + y^2}$$

$$= \frac{x(0)^2}{x^2 + (0)^2} = \frac{0}{x^2} = 0$$

along y-axis, $x = 0$

$$\lim_{(0,y)} \frac{xy^2}{x^2 + y^2}$$

$$= \frac{(0)\, y^2}{(0)^2 + y^2} = \frac{0}{y^2} = 0$$

another line, $y = x$

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$$\lim_{(x,y)\to(0,0)} \frac{x^2 y^2}{x^4 + y^4}$$

$$\lim_{(x,y)\to(0,0)} \frac{x^2(x^2)}{x^4 + (x)^4} = \frac{\cancel{x^4}}{2\cancel{x^4}} = \boxed{\frac{1}{2}}$$

limit doesn't exist, b/c limit of f(x,y) is different at different lines.